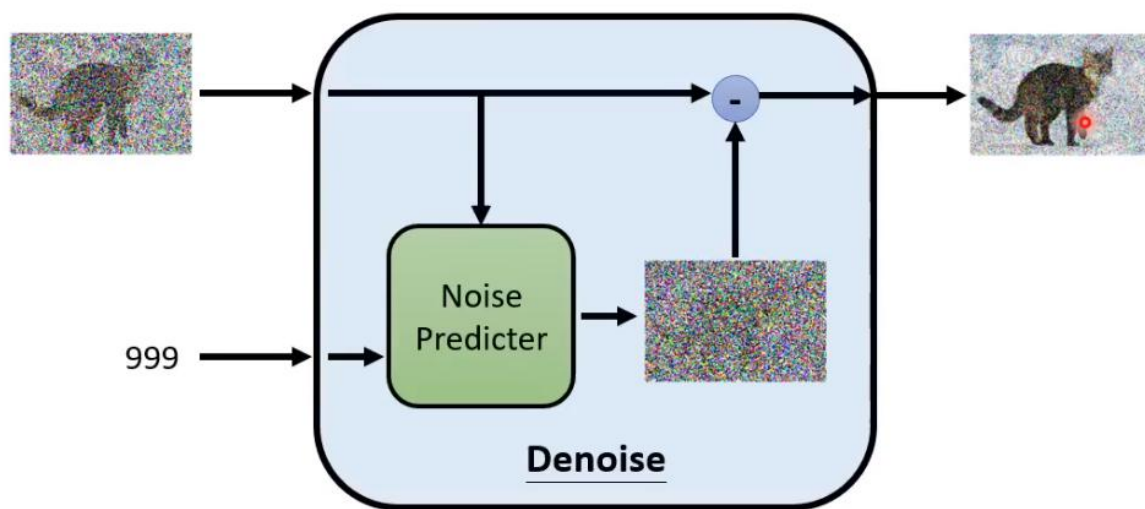
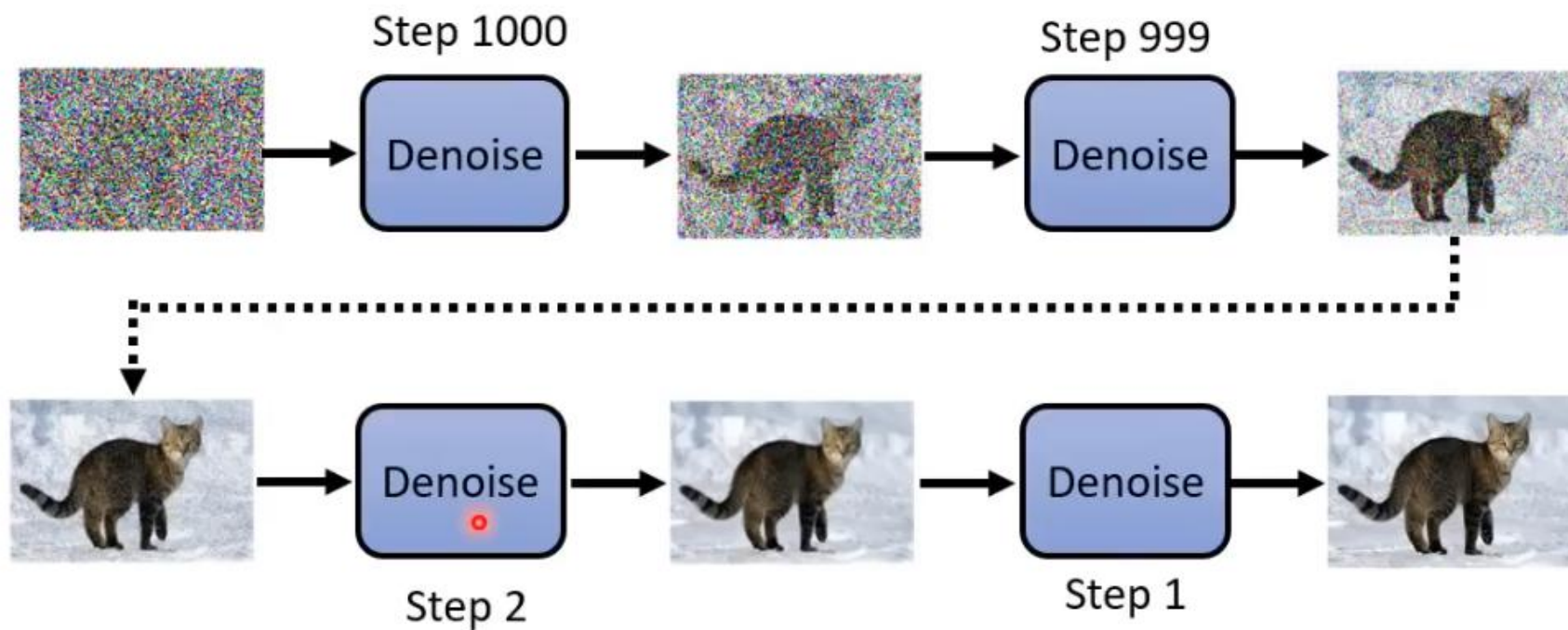


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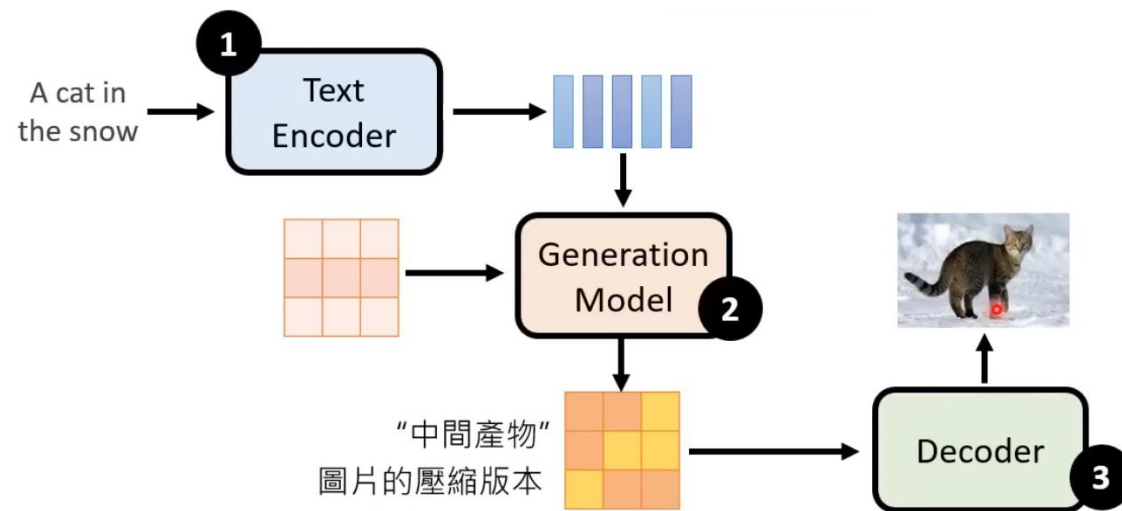
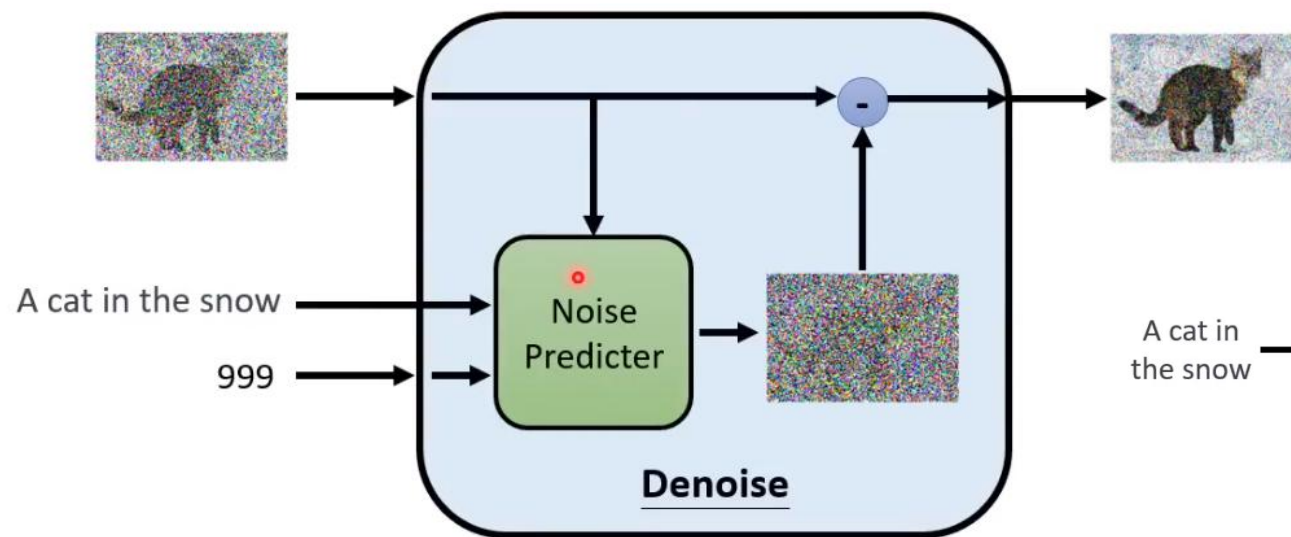
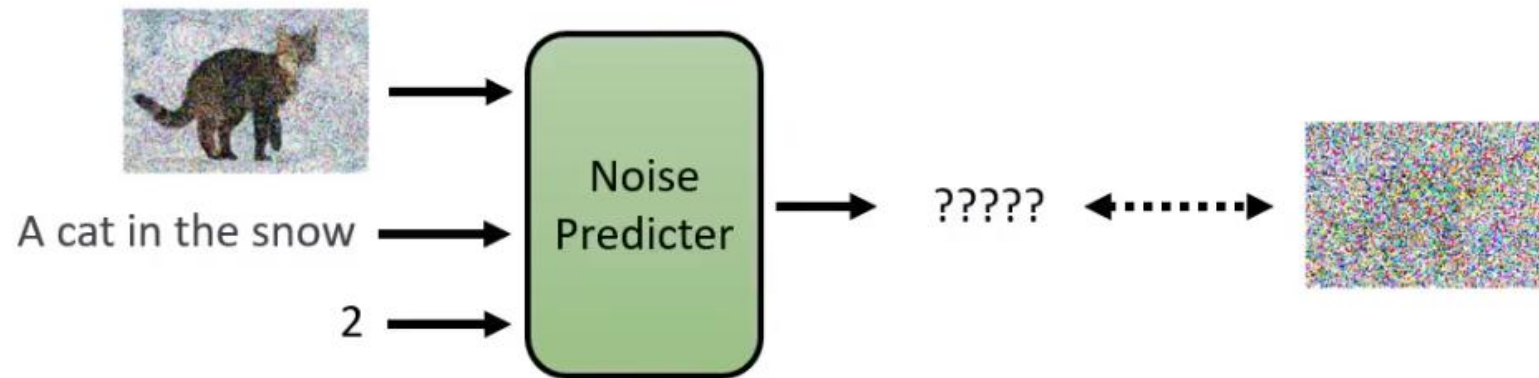
diffusion



带噪声图+加噪step ---Noise Predictor---> 噪声
带噪声图-噪声 -----> 新带噪声图（噪声更少）

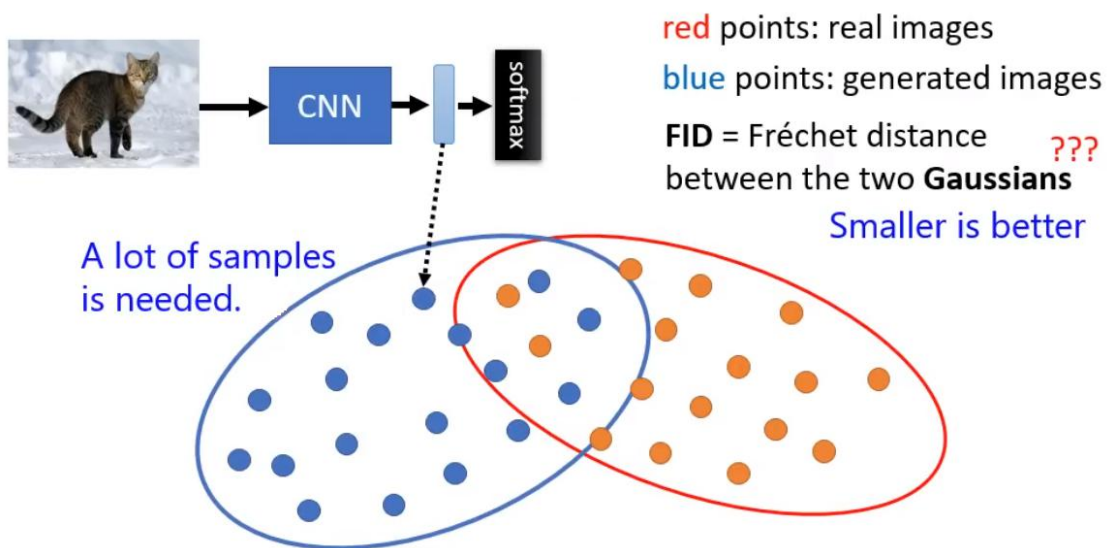
Net的唯一作用就是预测噪声

diffusion



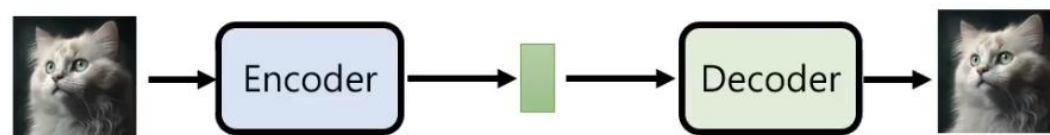
difussion

FID : FID (Fréchet Inception Distance) 是一种用于评估生成模型和真实数据分布之间差异的指标。

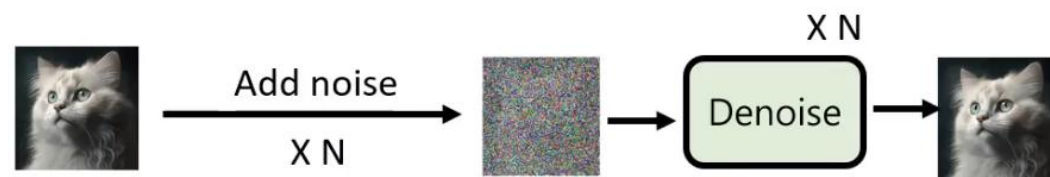


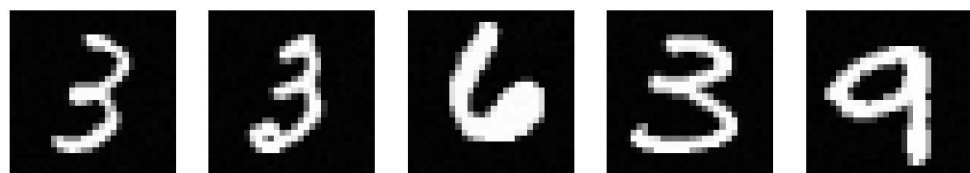
Constrastive Language-Image Pre-training : 采用对比学习的文本-图像预训练模型

VAE

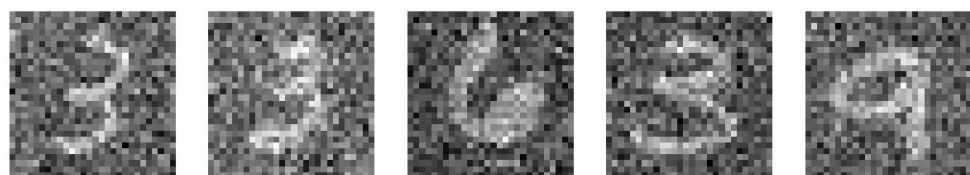


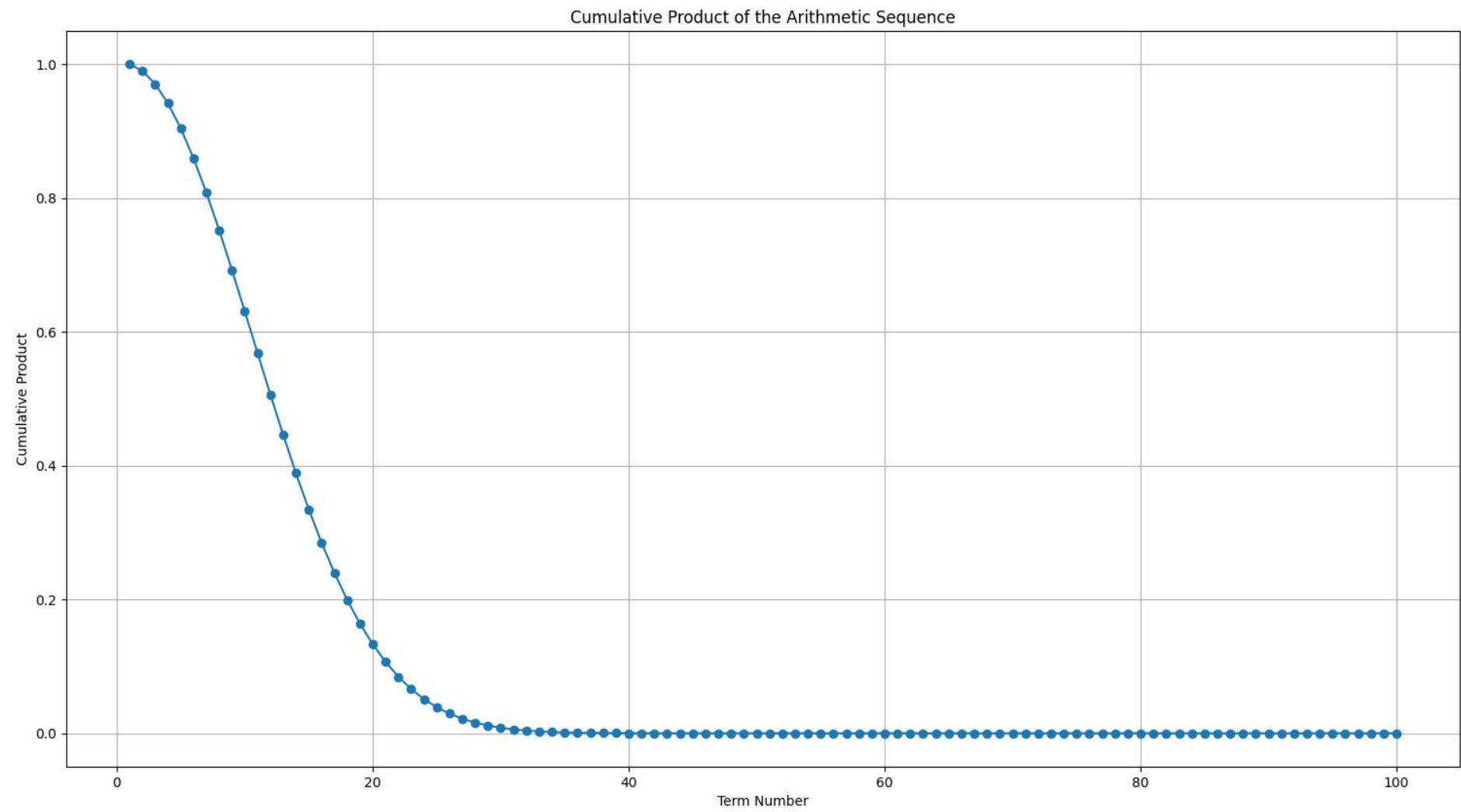
Diffusion



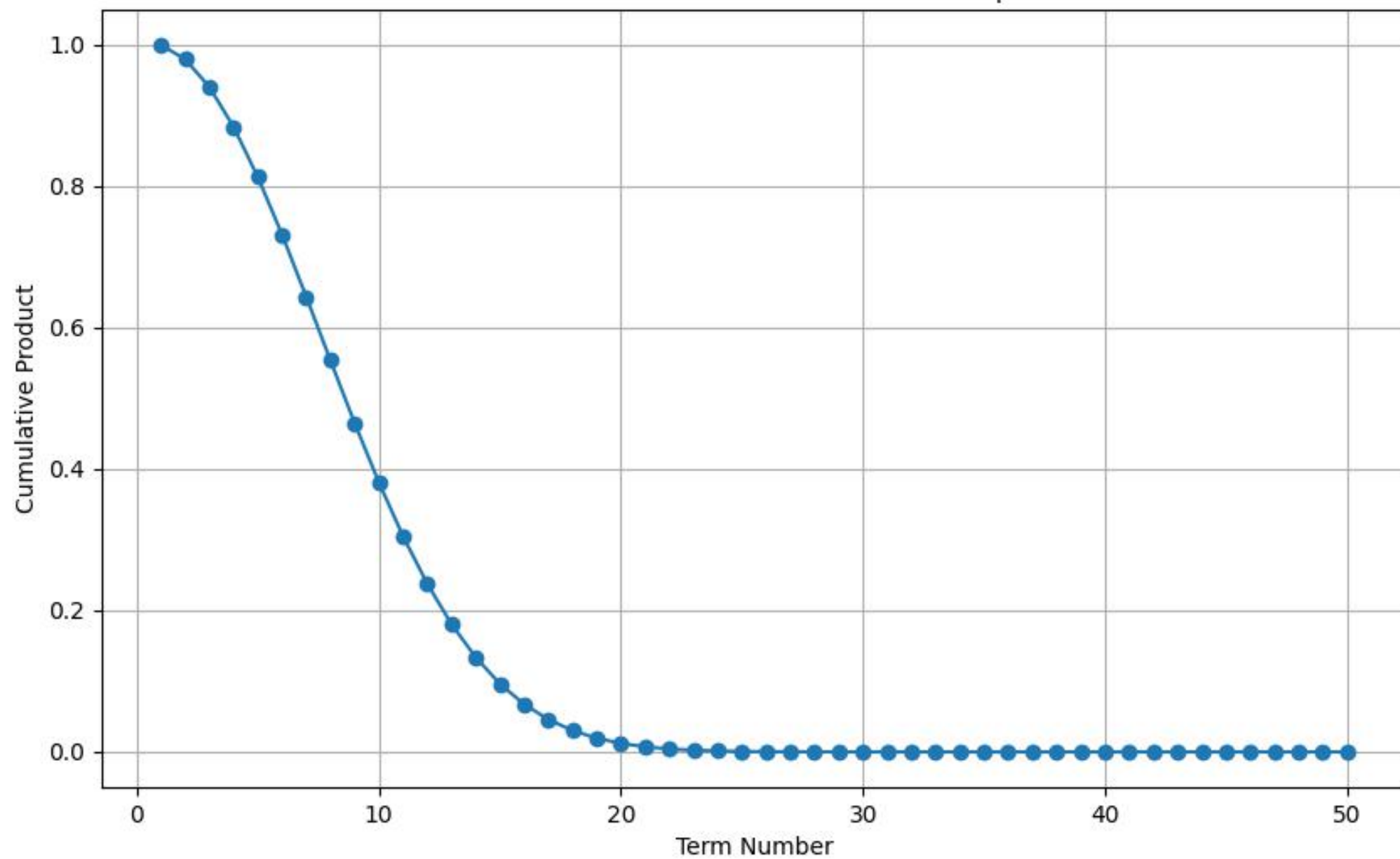


β_t 从 $\beta_1 = 10^{-4}$ 到 $\beta_T = 0.02$ 线性增长。





Cumulative Product of the Arithmetic Sequence



Algorithm 1 Training

```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
        $\nabla_{\theta} \left\| \epsilon - \epsilon_{\theta}(\underbrace{\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon}_{\text{加噪后新样本图}}, \underbrace{t}_{\text{时间步}}) \right\|^2$   
6: until converged
```

- 1.
2. 从训练集里取一个数据 $\mathbf{x}_0$
3. 随机选取一个时间步
4. 随机生成噪声 ϵ
5. 计算真实噪声 ϵ 和预测噪声 ϵ_{θ} ，预测值由时间步和加噪新图输入Net获得。优化真实噪声和预测噪声损失

Algorithm 2 Sampling

```

1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 

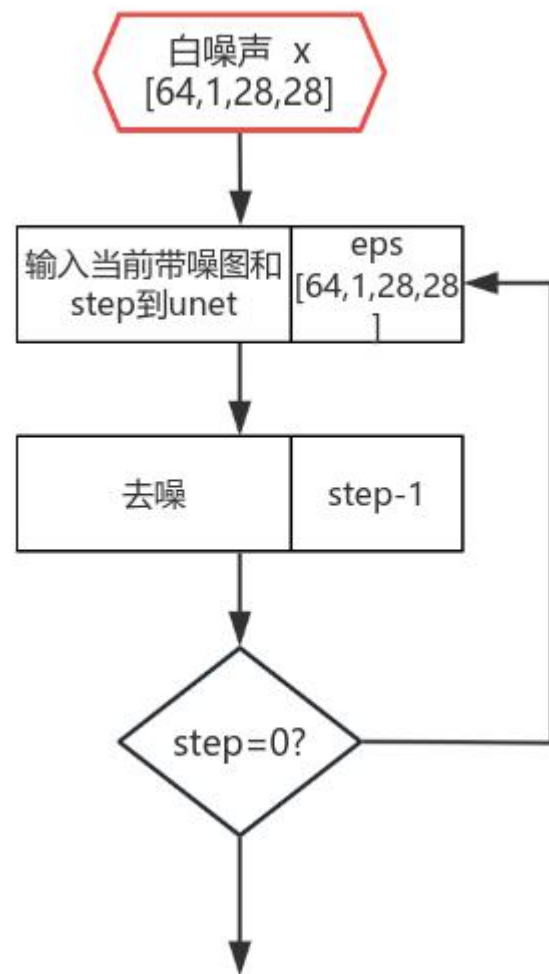
```

1. 生成纯噪声图 相当于加噪 T步
2. $t = T, T-1, \dots, 2, 1$
3. (t在0-1步加噪很特殊)
4. 一步一步按照公式去噪, 关键在于噪声预测
- 5.

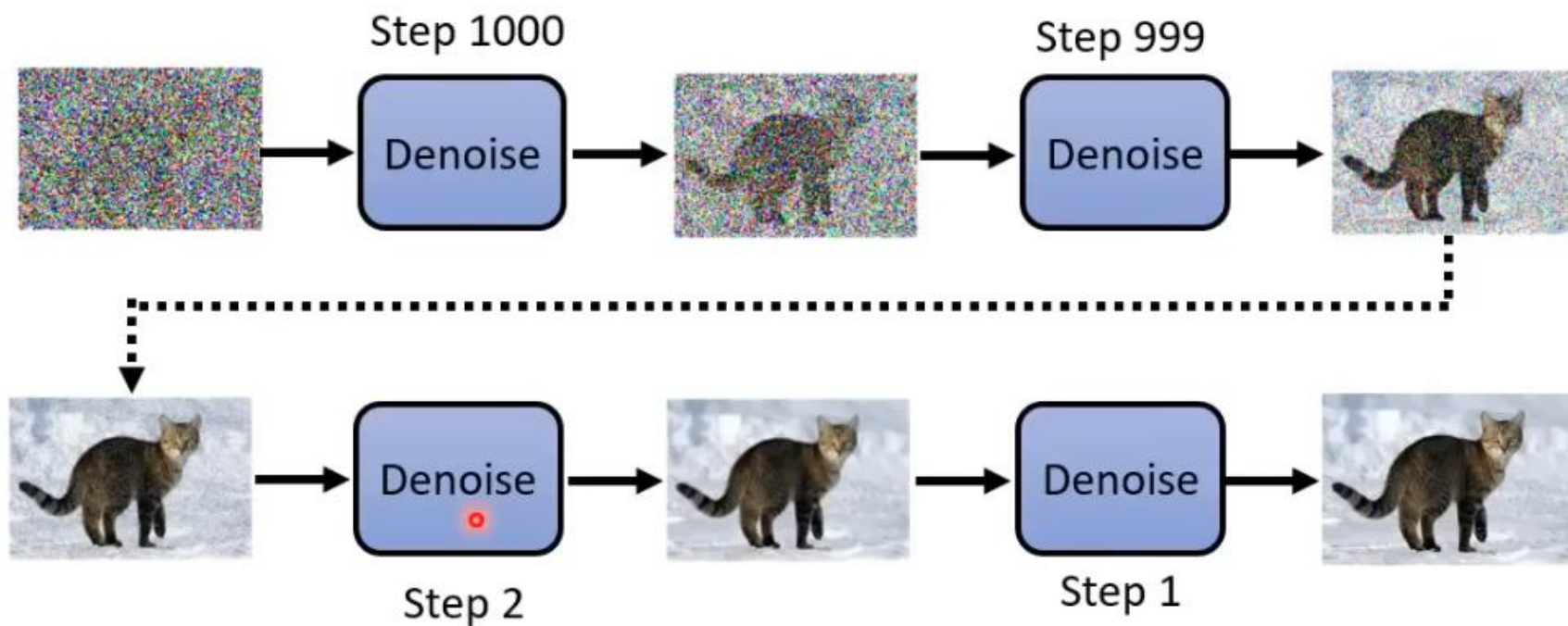
$$P(x_{t-1}|x_t) = N(x_{t-1}; \frac{1}{\sqrt{\alpha_t}} (x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon), \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t})$$

$$u = \frac{1}{\sqrt{\alpha_t}} (x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon)$$

$$\sigma^2 = \frac{\beta_t(1-\alpha_{t-1})}{1-\bar{\alpha}_t}$$



DDIM



$$q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0, \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \mathbf{I})$$

其中，只有参数 β_t 是可调的。 $\bar{\alpha}_t$ 是根据 β_t 算出的变量

$$\alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{i=1}^t \alpha_{i0}$$

$$q(x_{t-1}|x_t, x_0) = N(x_{t-1};$$

$$\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x_0,$$

$$\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t I)$$

$$\widetilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \qquad \frac{\widetilde{\beta}_t}{\beta_t} = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}$$

$$q(x_{t-1}|x_t, x_0) = N(x_{t-1};$$

$$\sqrt{\bar{\alpha}_{t-1}}x_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \widetilde{\beta}_t} \cdot \frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1 - \bar{\alpha}_t}},$$

$$\widetilde{\beta}_t I)$$

$$q(x_{t-1}|x_t, x_0) = N(x_{t-1};$$

$$\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x_0$$

$$\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \bullet \beta_t I$$

$$q(x_{t-1}|x_t, x_0) = N(x_{t-1} ;$$

$$\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x_0,$$

$$\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \bullet \beta_t I$$

$$q(x_{t-1}|x_t, x_0) = N(x_{t-1};$$

$$\sqrt{\bar{\alpha}_{t-1}}x_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \widetilde{\beta}_t} \bullet \frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1 - \bar{\alpha}_t}},$$

$$\widetilde{\beta}_t I$$